MSG No. 139.532 $I4/mmm1'$ [Type II, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: 4/mmm1'

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,33,34,35,36,37,38,39,40, 41,42,43,44,45,46,47,48]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,49,50,51,52,53,54,55,56, 57,58,59,60,61,62,63,64]

Table 2: Wyckoff site: 2b, site symmetry: 4/mmm1'

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,33,34,35,36,37,38,39,40, 41,42,43,44,45,46,47,48]
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,49,50,51,52,53,54,55,56, 57,58,59,60,61,62,63,64]

Table 3: Wyckoff site: 4c, site symmetry: mmm.1'

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	[1,4,5,6,9,12,13,14,33,36,37,38,41,44,45,46]
2	$\left[\frac{1}{2}, 0, 0\right]$	[2,3,7,8,10,11,15,16,34,35,39,40,42,43,47,48]
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[17,20,21,22,25,28,29,30,49,52,53,54,57,60,61,62]
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[18,19,23,24,26,27,31,32,50,51,55,56,58,59,63,64]

Table 4: Wyckoff site: 4d, site symmetry: -4m21'

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[1,6,12,13,23,24,26,27,33,38,44,45,55,56,58,59]
2	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[2,3,15,16,20,21,25,30,34,35,47,48,52,53,57,62]
3	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[4,5,9,14,18,19,31,32,36,37,41,46,50,51,63,64]
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[7,8,10,11,17,22,28,29,39,40,42,43,49,54,60,61]

Table 5: Wyckoff site: 4e, site symmetry: 4mm1'

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,6,12,13,15,16,33,34,35,38,44,45,47,48]
2	$[0, 0, -z]$	[4,5,7,8,9,10,11,14,36,37,39,40,41,42,43,46]
3	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	[17,18,19,22,28,29,31,32,49,50,51,54,60,61,63,64]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	[20,21,23,24,25,26,27,30,52,53,55,56,57,58,59,62]

Table 6: Wyckoff site: `8f`, site symmetry: `..2/m1'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,16,24,25,33,48,56,57]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,12,20,26,34,44,52,58]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[3,13,21,27,35,45,53,59]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[4,10,18,28,36,42,50,60]`
5	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[5,11,19,29,37,43,51,61]`
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[6,15,23,30,38,47,55,62]`
7	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[7,14,22,31,39,46,54,63]`
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[8,9,17,32,40,41,49,64]`

Table 7: Wyckoff site: `8g`, site symmetry: `2mm.1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,6,12,13,33,38,44,45]`
2	$\left[\frac{1}{2}, 0, z\right]$	`[2,3,15,16,34,35,47,48]`
3	$\left[0, \frac{1}{2}, -z\right]$	`[4,5,9,14,36,37,41,46]`
4	$\left[\frac{1}{2}, 0, -z\right]$	`[7,8,10,11,39,40,42,43]`
5	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	`[17,22,28,29,49,54,60,61]`
6	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	`[18,19,31,32,50,51,63,64]`
7	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	`[20,21,25,30,52,53,57,62]`
8	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	`[23,24,26,27,55,56,58,59]`

Table 8: Wyckoff site: `8h`, site symmetry: `m.2m1'`

No.	position	mapping
1	$[x, x, 0]$	`[1,7,14,16,33,39,46,48]`
2	$[-x, x, 0]$	`[2,5,11,12,34,37,43,44]`
3	$[x, -x, 0]$	`[3,4,10,13,35,36,42,45]`
4	$[-x, -x, 0]$	`[6,8,9,15,38,40,41,47]`
5	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	`[17,23,30,32,49,55,62,64]`
6	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[18,21,27,28,50,53,59,60]`
7	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	`[19,20,26,29,51,52,58,61]`
8	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}\right]$	`[22,24,25,31,54,56,57,63]`

Table 9: Wyckoff site: `8i`, site symmetry: `m2m.1'`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4,13,14,33,36,45,46]`
2	$[0, x, 0]$	`[2,7,11,16,34,39,43,48]`
3	$[0, -x, 0]$	`[3,8,10,15,35,40,42,47]`
4	$[-x, 0, 0]$	`[5,6,9,12,37,38,41,44]`
5	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,20,29,30,49,52,61,62]`
6	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	`[18,23,27,32,50,55,59,64]`
7	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	`[19,24,26,31,51,56,58,63]`

continued ...

Table 9

No.	position	mapping
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[21,22,25,28,53,54,57,60]

Table 10: Wyckoff site: 8j, site symmetry: m2m.1'

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	[1,4,13,14,33,36,45,46]
2	$\left[\frac{1}{2},\ x,\ 0\right]$	[2,7,11,16,34,39,43,48]
3	$\left[\frac{1}{2},\ -x,\ 0\right]$	[3,8,10,15,35,40,42,47]
4	$\left[-x,\ \frac{1}{2},\ 0\right]$	[5,6,9,12,37,38,41,44]
5	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[17,20,29,30,49,52,61,62]
6	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[18,23,27,32,50,55,59,64]
7	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[19,24,26,31,51,56,58,63]
8	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[21,22,25,28,53,54,57,60]

Table 11: Wyckoff site: 16k, site symmetry: ..21'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[1,23,33,55]
2	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	[2,21,34,53]
3	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[3,20,35,52]
4	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[4,19,36,51]
5	$\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[5,18,37,50]
6	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[6,24,38,56]
7	$\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$	[7,17,39,49]
8	$\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$	[8,22,40,54]
9	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[9,31,41,63]
10	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[10,29,42,61]
11	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[11,28,43,60]
12	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[12,27,44,59]
13	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[13,26,45,58]
14	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[14,32,46,64]
15	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$	[15,25,47,57]
16	$\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[16,30,48,62]

Table 12: Wyckoff site: 16l, site symmetry: m..1'

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,14,33,46]
2	$[-y,\ x,\ 0]$	[2,11,34,43]
3	$[y,\ -x,\ 0]$	[3,10,35,42]
4	$[x,\ -y,\ 0]$	[4,13,36,45]
5	$[-x,\ y,\ 0]$	[5,12,37,44]
6	$[-x,\ -y,\ 0]$	[6,9,38,41]

continued ...

Table 12

No.	position	mapping
7	$[y,\ x,\ 0]$	`[7,16,39,48]`
8	$[-y,\ -x,\ 0]$	`[8,15,40,47]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[17,30,49,62]`
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[18,27,50,59]`
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[19,26,51,58]`
12	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[20,29,52,61]`
13	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[21,28,53,60]`
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[22,25,54,57]`
15	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[23,32,55,64]`
16	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[24,31,56,63]`

Table 13: Wyckoff site: `16m`, site symmetry: `..m1'`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,16,33,48]`
2	$[-x,\ x,\ z]$	`[2,12,34,44]`
3	$[x,\ -x,\ z]$	`[3,13,35,45]`
4	$[x,\ -x,\ -z]$	`[4,10,36,42]`
5	$[-x,\ x,\ -z]$	`[5,11,37,43]`
6	$[-x,\ -x,\ z]$	`[6,15,38,47]`
7	$[x,\ x,\ -z]$	`[7,14,39,46]`
8	$[-x,\ -x,\ -z]$	`[8,9,40,41]`
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[17,32,49,64]`
10	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[18,28,50,60]`
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[19,29,51,61]`
12	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[20,26,52,58]`
13	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[21,27,53,59]`
14	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[22,31,54,63]`
15	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[23,30,55,62]`
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[24,25,56,57]`

Table 14: Wyckoff site: `16n`, site symmetry: `.m.1'`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,12,33,44]`
2	$[-y,\ 0,\ z]$	`[2,15,34,47]`
3	$[y,\ 0,\ z]$	`[3,16,35,48]`
4	$[0,\ -y,\ -z]$	`[4,9,36,41]`
5	$[0,\ y,\ -z]$	`[5,14,37,46]`
6	$[0,\ -y,\ z]$	`[6,13,38,45]`
7	$[y,\ 0,\ -z]$	`[7,10,39,42]`
8	$[-y,\ 0,\ -z]$	`[8,11,40,43]`
9	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[17,28,49,60]`
10	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[18,31,50,63]`

continued ...

Table 14

No.	position	mapping
11	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[19,32,51,64]
12	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[20,25,52,57]
13	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[21,30,53,62]
14	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[22,29,54,61]
15	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[23,26,55,58]
16	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[24,27,56,59]

Table 15: Wyckoff site: 32o, site symmetry: 11'

No.	position	mapping
1	$[x,\ y,\ z]$	[1,33]
2	$[-y,\ x,\ z]$	[2,34]
3	$[y,\ -x,\ z]$	[3,35]
4	$[x,\ -y,\ -z]$	[4,36]
5	$[-x,\ y,\ -z]$	[5,37]
6	$[-x,\ -y,\ z]$	[6,38]
7	$[y,\ x,\ -z]$	[7,39]
8	$[-y,\ -x,\ -z]$	[8,40]
9	$[-x,\ -y,\ -z]$	[9,41]
10	$[y,\ -x,\ -z]$	[10,42]
11	$[-y,\ x,\ -z]$	[11,43]
12	$[-x,\ y,\ z]$	[12,44]
13	$[x,\ -y,\ z]$	[13,45]
14	$[x,\ y,\ -z]$	[14,46]
15	$[-y,\ -x,\ z]$	[15,47]
16	$[y,\ x,\ z]$	[16,48]
17	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17,49]
18	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[18,50]
19	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[19,51]
20	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[20,52]
21	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[21,53]
22	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[22,54]
23	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[23,55]
24	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[24,56]
25	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[25,57]
26	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[26,58]
27	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[27,59]
28	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[28,60]
29	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[29,61]
30	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[30,62]
31	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[31,63]
32	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[32,64]